

2(a)(i)	any four from: 1 glycolysis ; 2 glucose to pyruvate ; 3 pyruvate to ethanal by, decarboxylation / CO₂ removal ; 4 ethanal, reduced / hydrogenated, to ethanol ; 5 reduced NAD to NAD ; A regeneration of NAD 6 AVP ; e.g. ethanol / alcohol, dehydrogenase or pyruvate decarboxylase	4
2(a)(ii)	any two from: 1 can survive (in absence of oxygen) ; 2 (as) ATP still produced (from glycolysis) ; 3 glycolysis can continue ; 4 as NAD recycled ;	2
2(b)	any two from: 1 GM strain has higher (concentration of ethanol / ethanol production) or GM strain has steeper increase (in concentration of ethanol / ethanol production) ; ora 2 (both GM and non-GM) (concentration of ethanol / ethanol production) increases then levels off ; 3 comparative figs to support trend ; e.g. (mp1) (max concentration) for GM 19 g dm⁻³ and non-GM 1 g dm⁻³ (mp2) (levels off at) 18 g dm⁻³ for GM and 1 g dm⁻³ for non-GM	2

Question	Answer	Marks
2(c)	any two from: 1 similar / same, concentration of starch hydrolysed ; 2 similar / same, concentration of glucose available (to yeast) ; 3 enzymes work at the same rate (in both methods) ; 4 AVP ; e.g. GM yeast cells produce more enzymes	2
2(d)	any one from: 1 more (named) fuel or electricity needed ; 2 need to add enzymes ; 3 AVP ; e.g. (whole method) takes longer	1